Project Structure

```
mern-redux-counter
│
├── backend
│   └── server.js
│
└── frontend
    ├── src
    │   ├── app
    │   │   └── store.js
    │   ├── features
    │   │   └── counterSlice.js
    │   ├── components
    │   │   └── Counter.jsx
    │   ├── App.jsx
    │   └── main.jsx
    │
    └── package.json
```

Backend (Node + Express)

Create folder **backend**

Install packages

```
npm init -y
npm install express cors
```

server.js

```
const express = require("express")
const cors = require("cors")

const app = express()

app.use(cors())
app.use(express.json())

let counter = 0

// get counter value
app.get("/counter", (req, res) => {
   res.json({ value: counter })
})

// increment counter
app.post("/increment", (req, res) => {
   counter++
   res.json({ value: counter })
```

```
})

// decrement counter
app.post("/decrement", (req, res) => {
   counter--
   res.json({ value: counter })
})

app.listen(5000, () => {
   console.log("Server running on port 5000")
})
```

Run backend

```
node server.js
```

Server runs on

http://localhost:5000

Create Vite React Frontend

```
npm create vite@latest frontend
```

Choose

React
JavaScript

Then install dependencies

```
cd frontend
npm install
```

Install Redux + Axios

```
npm install @reduxjs/toolkit react-redux axios
```

Redux Store

Create folder

src/app

store.js

```
import { configureStore } from "@reduxjs/toolkit"
import counterReducer from "../features/counterSlice"
```

```
export const store = configureStore({
  reducer: {
    counter: counterReducer
  }
})
```

Redux Slice

Create folder

src/features

counterSlice.js

```
import { createSlice, createAsyncThunk } from "@reduxjs/toolkit"
import axios from "axios"

const API = "http://localhost:5000"

// get counter from backend
export const fetchCounter = createAsyncThunk(
  "counter/fetchCounter",
  async () => {
    const res = await axios.get(`${API}/counter`)
    return res.data.value
  }
)

// increment
export const incrementCounter = createAsyncThunk(
  "counter/incrementCounter",
  async () => {
    const res = await axios.post(`${API}/increment`)
    return res.data.value
  }
)

// decrement
export const decrementCounter = createAsyncThunk(
  "counter/decrementCounter",
  async () => {
    const res = await axios.post(`${API}/decrement`)
    return res.data.value
  }
)

const counterSlice = createSlice({
  name: "counter",
  initialState: {
    value: 0
```

```
  },
  reducers: {},
  extraReducers: (builder) => {
    builder
      .addCase(fetchCounter.fulfilled, (state, action) => {
        state.value = action.payload
      })
      .addCase(incrementCounter.fulfilled, (state, action) => {
        state.value = action.payload
      })
      .addCase(decrementCounter.fulfilled, (state, action) => {
        state.value = action.payload
      })
  }
})

export default counterSlice.reducer
```

Connect Redux with React

main.jsx

```
import React from "react"
import ReactDOM from "react-dom/client"
import App from "./App.jsx"
import { Provider } from "react-redux"
import { store } from "./app/store"

ReactDOM.createRoot(document.getElementById("root")).render(
  <React.StrictMode>
    <Provider store={store}>
      <App />
    </Provider>
  </React.StrictMode>
)
```

Provider makes Redux store available to React components.

Counter Component

Create folder

src/components

Counter.jsx

```
import React, { useEffect } from "react"
import { useSelector, useDispatch } from "react-redux"
```

```
import {
  fetchCounter,
  incrementCounter,
  decrementCounter
} from "../features/counterSlice"

function Counter() {

  const count = useSelector((state) => state.counter.value)

  const dispatch = useDispatch()

  useEffect(() => {
    dispatch(fetchCounter())
  }, [])

  return (
    <div>

      <h2>Counter: {count}</h2>

      <button onClick={() => dispatch(incrementCounter())}>
        Increment
      </button>

      <button onClick={() => dispatch(decrementCounter())}>
        Decrement
      </button>

    </div>
  )
}

export default Counter
```

App Component

App.jsx

```
import Counter from "./components/Counter"

function App() {

  return (
    <div>

      <h1>MERN Redux Counter App</h1>

      <Counter/>
```

```
    </div>
  )

}

export default App
```

Run the Project

Run backend

```
cd backend
node server.js
```

Run frontend

```
cd frontend
npm run dev
```

Vite will start server like

http://localhost:5173

```
Button Click
    ↓
Dispatch Action
    ↓
Redux Thunk calls API
    ↓
Backend updates counter
    ↓
Reducer updates state
    ↓
React Component re-renders
```